

HR Analytics Statistical Analysis Report

Descriptive & Inferential Analysis of Employee Data
By Adrij Das

1. Dataset Overview

This report presents a statistical analysis of a cleaned human resources dataset containing records for 20,000 employees across multiple European countries. The dataset includes demographic, job-related, performance, and compensation variables. The analysis covers descriptive statistics, visual exploration, and inferential tests relevant to workforce planning and compensation strategy.

Metric	Value	Metric	Value
Total Employees	20,000	Departments	5 (HR, Ops, IT, Sales, Finance)
Active Employees	86.8%	Countries	7 (UK, NL, PL, FR, IT, DE, ES)
Resigned	11.0%	Work Modes	On-site, Hybrid, Remote
Terminated	2.2%	Job Levels	Junior, Mid, Senior, Director

2. Descriptive Statistics

Summary statistics for the primary numeric variables. Salary exhibits moderate right-skew (skew ~ 0.57), while Age, Experience, and Tenure are approximately symmetric. Performance Score is slightly left-skewed, indicating a tendency toward higher ratings.

Variable	Mean	Median	Std Dev	Min	Max	Skew	Kurtosis
Salary (EUR)	83,388	80,981	34,296	35,000	280,000	0.57	0.13
Age (years)	39.76	40.0	10.29	21	59	0.03	-0.79
Experience (yrs)	17.15	17.0	9.84	0	35	0.02	-0.85
Tenure (yrs)	9.33	9.3	5.32	0.2	18.6	0.01	-1.20
Perf. Score	2.64	3.0	0.84	1	4	-0.22	-0.51

Table 1. Descriptive statistics for key numeric variables.

Salary by Department: Sales has the highest average (EUR 87,571), followed by HR (EUR 84,791). Finance has the lowest (EUR 80,115). Differences are modest relative to within-group variation.

Performance Ratings: Good (45.4%), Satisfactory (30.7%), Excellent (14.3%), Needs Improvement (9.7%). Most employees fall into the middle-to-upper performance bands.

3. Visual Exploration

3.1 Salary Distribution

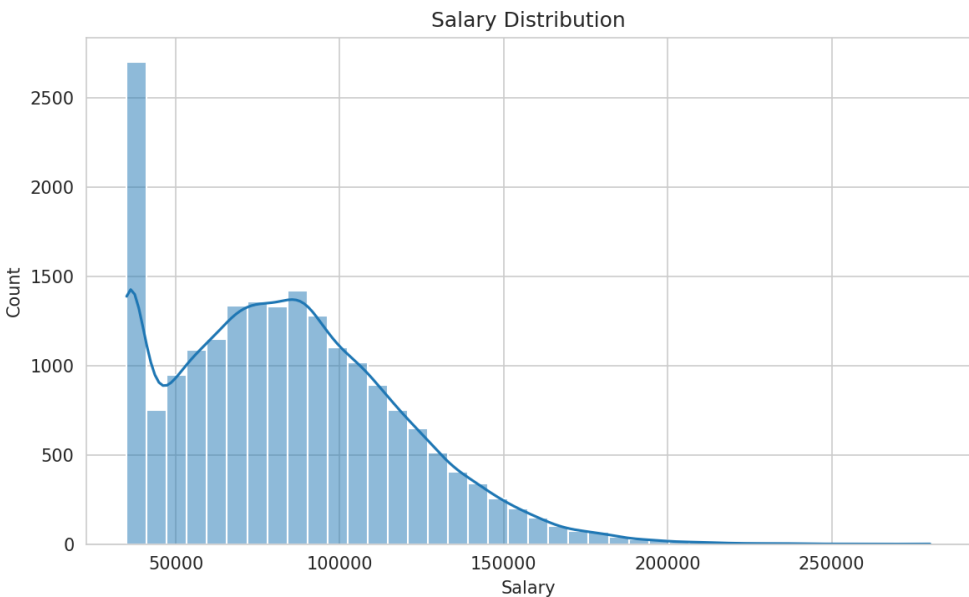


Figure 1. Distribution of employee salaries with kernel density estimate. Moderate right skew is visible.

3.2 Salary by Department

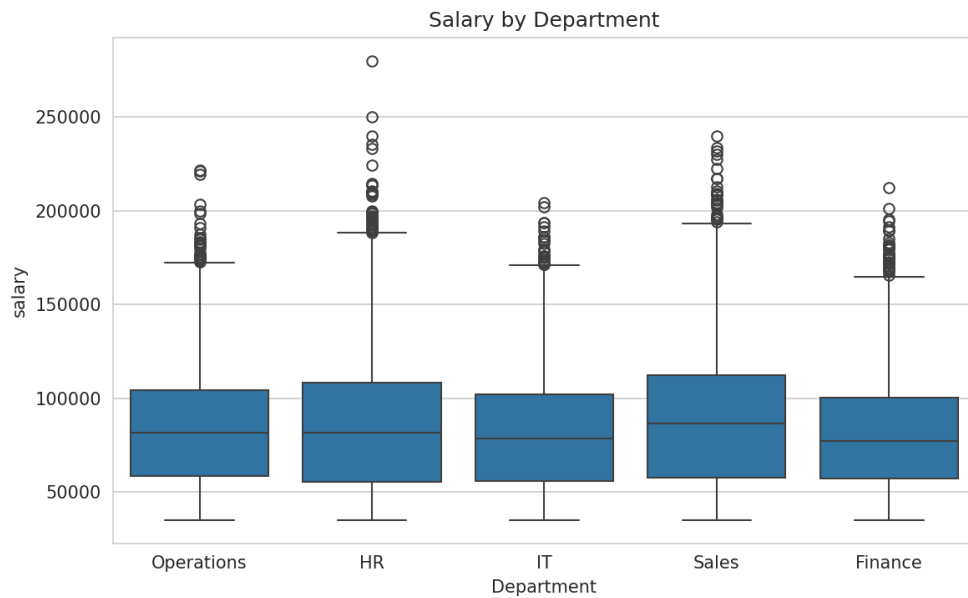


Figure 2. Boxplots of salary by department. Medians and spreads are broadly comparable across departments.

3.3 Average Performance Score by Department

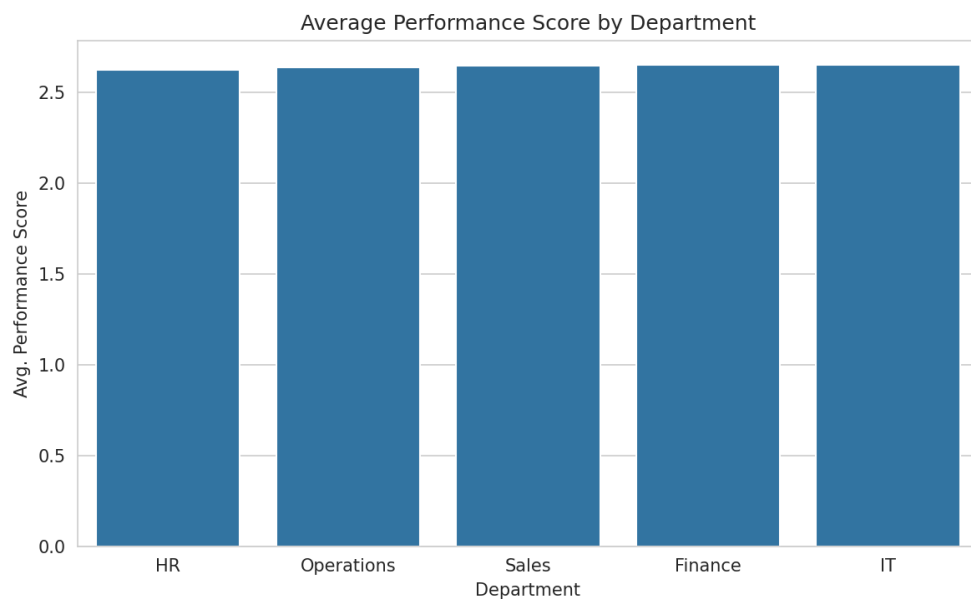


Figure 3. Mean performance score by department. Scores are nearly identical across all departments (~2.62–2.65).

3.4 Salary vs. Experience



Figure 4. Scatterplot of salary against years of experience (2,000-point sample). No clear linear relationship is apparent.

3.5 Correlation Heatmap

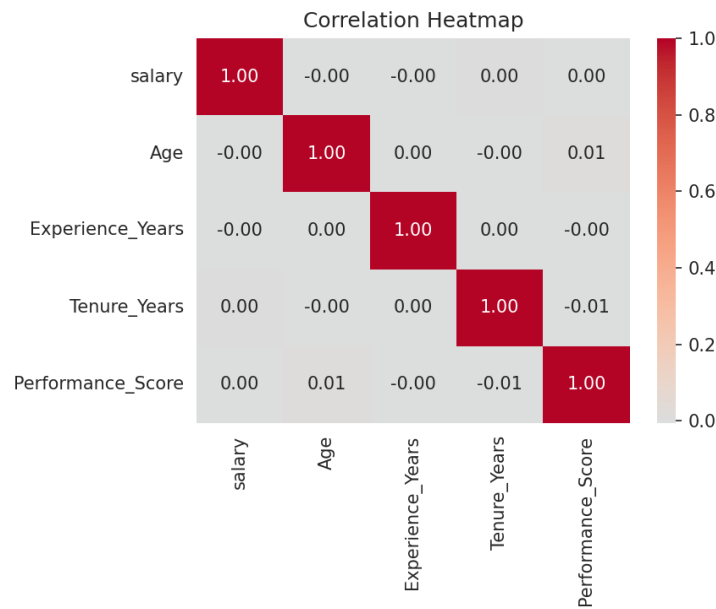


Figure 5. Pearson correlation matrix among numeric variables. Correlations are near zero for most pairs.

3.6 Performance Rating Distribution



Figure 6. Counts of employees by performance rating category.

3.7 Employee Status Breakdown

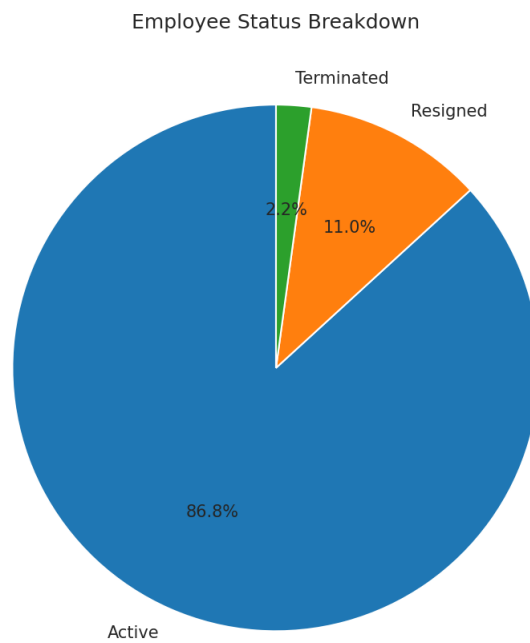


Figure 7. Proportion of employees by employment status (Active, Resigned, Terminated).

3.8 Hiring Trend by Year

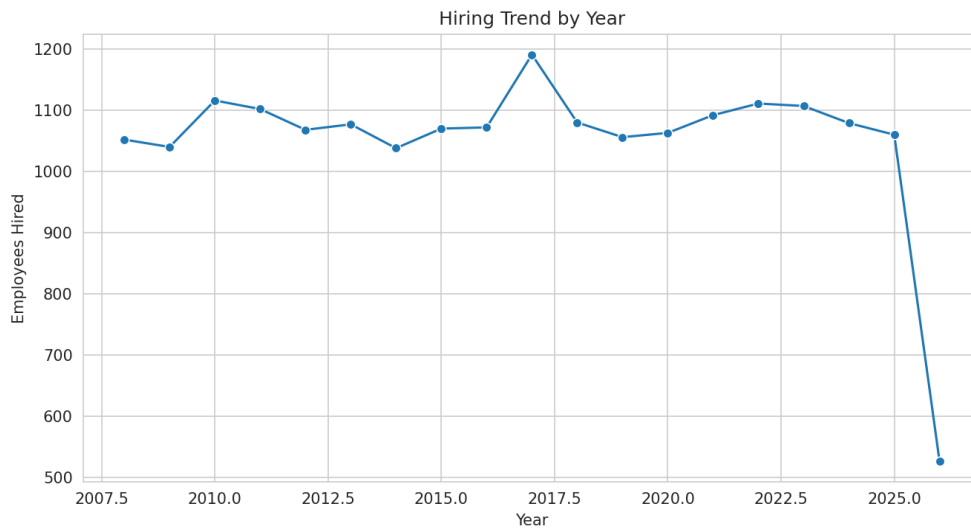


Figure 8. Number of employees hired per year. Hiring volume has been relatively stable with some year-to-year fluctuation.

4. Inferential Statistical Analysis

All hypothesis tests use a significance level of $\alpha = 0.05$. Results are summarized below.

4.1 One-Way ANOVA: Performance Score by Department

Null hypothesis (H_0): Mean performance scores are equal across the five departments. Result: $F = 0.93$, $p = 0.444$. Fail to reject H_0 . There is no statistically significant difference in average performance scores across departments.

4.2 One-Way ANOVA: Salary by Job Level

Null hypothesis (H_0): Mean salaries are equal across job levels (Junior, Mid, Senior, Director). Result: $F = 40.23$, $p = 6.6e-26$. Reject H_0 . Mean salaries differ significantly by job level.

Job Level	Mean Salary (EUR)	Interpretation
Director	86,355	Highest average
Mid	86,071	Near Director level
Senior	81,018	Moderate
Junior	80,605	Lowest average

Table 2. Mean salary by job level. Note: Mid-level mean is close to Director, suggesting possible compression or band overlap.

4.3 Pearson Correlation: Salary vs. Experience

Result: $r = -0.002$, $p = 0.825$. No significant linear relationship between salary and years of experience. Experience alone does not predict compensation in this dataset.

4.4 Pearson Correlation: Tenure vs. Performance Score

Result: $r = -0.006$, $p = 0.432$. No significant linear relationship between tenure and performance score.

4.5 Welch's t-test: Salary – Remote vs. On-site

Comparing mean salaries of Remote ($n = 4,991$) and On-site ($n = 9,529$) employees. Result: $t = 1.57$, $p = 0.116$. Fail to reject H_0 . No statistically significant difference in mean salary between remote and on-site workers at $\alpha = 0.05$.

4.6 Chi-Square Test: Job Level vs. Employment Status

Tests association between job level and status (Active / Resigned / Terminated). Result: $\chi^2 = 8.45$, $p = 0.207$. Fail to reject H_0 . No significant association between job level and employment status.

5. Key Findings & Practical Implications

- Performance is uniform across departments.** ANOVA showed no meaningful differences in average performance scores. Department-level performance management appears consistent.
- Job level drives salary differences.** The only strong structural salary difference is by job level (highly significant ANOVA). Director and Mid roles command the highest averages; Junior the lowest.
- Experience and tenure do not predict salary or performance.** Near-zero correlations suggest compensation and ratings are not strongly tied to time-based metrics in this organization.
- Work mode does not affect salary.** Remote and on-site employees have statistically similar average pay, supporting equitable pay practices across work arrangements.
- Workforce is stable.** 86.8% of employees are active; resignation rate is 11% and termination only 2.2%. Status is independent of job level.

- **Salary distribution is moderately right-skewed.** A minority of high earners pull the mean above the median (mean EUR 83.4k vs median EUR 81.0k).